

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

6. Project Testing Phase

The Project Testing Phase is performed to verify that the FitBuddy application works correctly and that each feature produces the expected result. Testing is important because the application contains multiple components including the frontend, backend, database, and Gemini AI integration.

First, the project structure and Python imports were tested. Each AI module was checked to ensure that the required functions could be imported correctly. This helped identify and correct module-location and function-name issues during development.

The FastAPI application was then tested by starting the server using Uvicorn. The application health endpoint was used to verify whether the backend was running correctly.

The user-input functionality was tested using different combinations of User ID, name, age, weight, fitness goal, and workout intensity. Invalid or duplicate User IDs were also checked to ensure that the application handles errors correctly.

The workout-generation feature was tested to verify that a 7-day workout plan is produced from the supplied user information. The nutrition and recovery recommendation was also checked.

Database testing was performed to verify that user information and generated plans were correctly stored. The original workout plan was checked to ensure that it could be retrieved later for feedback processing.

The feedback functionality was tested by entering feedback after generating an initial plan. The application was checked to ensure that the feedback was processed and that a revised workout plan was generated and stored.

API endpoints were also tested to verify correct responses and error handling. The demo mode was useful during testing because it allowed the application functionality to be tested without depending completely on an external Gemini API request.

The main testing areas were:

- Frontend form testing.
- Input validation testing.
- FastAPI route testing.
- Gemini integration testing.
- Workout generation testing.
- Nutrition recommendation testing.
- Database testing.

- Feedback testing.
- Updated-plan testing.
- Error handling testing.
- API testing.

Through these tests, errors were identified and corrected during development, ensuring that the major FitBuddy functions operate together correctly.